

Jagan K K: Introduction

Jagan K K is a motivated Computer Science graduate from St. Thomas College of Engineering and Technology with a strong passion for Artificial Intelligence, Machine Learning, Augmented Reality, and software development. With expertise in Python, C, Java, MySQL, Unity, Flask, and YOLO-based deep learning systems, he enjoys building innovative projects that combine creativity with real-world problem solving. His curiosity for emerging technologies drives him to continuously learn, experiment, and create impactful applications that improve user interaction and digital experiences.

He has worked on multiple projects that reflect both technical and creative abilities. His major project, *Memory Palace*, is an AR-powered learning system that uses YOLO-based object detection and Unity to overlay digital flashcards onto real-world locations for context-aware education. He also developed *SketchPlay AI*, a real-time augmented 2D game where hand-drawn sketches interact with gameplay using YOLOv8, RNN, and MediaPipe. In addition, he created *NetBackup Pro*, a Flask-based backup monitoring dashboard that automates system tracking and visualization across multiple devices using SSH, and *Ani-verse*, a 2D Metroidvania-style hack-and-slash game built in Unity with custom pixel-art assets.

Jagan K K: Creative Problem Solver and Technology Enthusiast

Beyond academics, Jagan is passionate about building systems that merge intelligence, interactivity, and immersive experiences. He enjoys exploring futuristic ideas through projects that integrate AI, AR, automation, and gaming technologies. His technical foundation is strengthened by hands-on experience during his DevOps internship at Kerala Vision Broadband Ltd.

Jagan K K : Education

St. Thomas College of Engineering and Technology Kannur, Kerala

Bachelor of Technology in Computer Science

CGPA:8.38/10

Indira Gandhi Public School

Higher Secondary Education (Science Stream)

- Plus Two: 79%
- SSLC: 78%

Jagan K K : Technical Skills

Languages : Python , C

Database System : MySQL

AI Frameworks : Langchain , Langgraph , RAG , Vectorless RAG

Jagan K K : Projects

Memory Palace (Major Project) | Unity, Python, HTML, CSS, YOLO

Developed an AR-based system to augment digital flashcards in real-world locations for context-aware learning.

Integrated YOLO-based object detection for recognizing specific boards/walls as anchors for overlaying information.

Enabled area-specific data retrieval, allowing personalized educational content to appear only in predefined physical spaces.

Built backend logic with Python and web components (HTML/CSS) for managing content and user interaction.

Purpose is to retrieving documents at ease without need for logging to any website

SketchPlay AI | Python, OpenCV, YOLO

Collaborated with a team to develop a real-time 2D augmented game driven by hand-drawn sketches.

Trained and implemented deep learning models including YOLOv8 and RNN for object detection and interaction.

Designed to run smoothly on medium-spec laptops at 30 FPS, with gameplay controlled via hand gestures using MediaPipe.

NetBackup Pro | Python, Flask, HTML, CSS, SSH

Developed a web-based dashboard to automate backup monitoring and retrieval of system information across multiple devices using SSH.

Implemented device tracking, backup status visualization, and historical data analysis for improved system reliability.

Built the interface with Flask, HTML, and CSS for real-time interaction and reporting.